

Exercise sheet 3

Rough Path Theory

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Problem 1

Recall that, for a rough path $\mathbf{X} = (X, \mathbb{X}) \in \mathcal{C}^\alpha$, the bracket of $\mathbf{X}$ is defined as the path $[\mathbf{X}]: [0, T] \rightarrow \mathbb{R}^{d \times d}$ given by

$$[\mathbf{X}]_t := X_{0,t} \otimes X_{0,t} - 2 \operatorname{Sym}(\mathbb{X}_{0,t}).$$

Show that

$$[\mathbf{X}]_{s,t} = X_{s,t} \otimes X_{s,t} - 2 \operatorname{Sym}(\mathbb{X}_{s,t})$$

for all $(s, t) \in \Delta_{[0,T]}$.

Problem 2

Let $\mathbf{X} = (X, \mathbb{X}) \in \mathcal{C}^\alpha$ be a rough path, and let $\Gamma \in \mathcal{C}^{2\alpha}$. Let $Z_t = X_t + \Gamma_t$ for $t \in [0, T]$, and

$$\mathbb{Z}_{s,t} = \mathbb{X}_{s,t} + \int_s^t X_{s,u} \otimes d\Gamma_u + \int_s^t \Gamma_{s,u} \otimes dX_u + \int_s^t \Gamma_{s,u} \otimes d\Gamma_u$$

for $(s, t) \in \Delta_{[0,T]}$, where the three integrals on the right-hand side are defined as Young integrals.

Part (a) Show that $\mathbf{Z} = (Z, \mathbb{Z})$ is a rough path.

Part (b) Show that $[\mathbf{Z}] = [\mathbf{X}]$.

Part (c) Deduce that $\mathbf{Z}$ is weakly geometric if and only if $\mathbf{X}$ is weakly geometric.

Problem 3

Let $\mathbf{X} = (X, \mathbb{X}) \in \mathcal{C}^\alpha$ be a rough path, and suppose that $(Y, Y') \in \mathcal{D}_X^{2\alpha}$ and $(Y', Y'') \in \mathcal{D}_X^{2\alpha}$ are controlled paths. Suppose further that

$$Y_t = Y_0 + \int_0^t Y'_s d\mathbf{X}_s + \Gamma_t$$

for all $t \in [0, T]$, for some path $\Gamma \in \mathcal{C}^{2\alpha}$. Let $f \in C^3$.

Prove that

$$f(Y_T) = f(Y_0) + \int_0^T Df(Y_u) Y'_u d\mathbf{X}_u + \int_0^T Df(Y_u) d\Gamma_u + \frac{1}{2} \int_0^T D^2 f(Y_u) (Y'_u \otimes Y'_u) d[\mathbf{X}]_u.$$

Problem 4

Suppose that $\mathbf{X} = (X, \mathbb{X}) \in \mathcal{C}^\alpha$ and $(K, K') \in \mathcal{D}_X^{2\alpha}$ are such that the rough integral $\int_0^\cdot K_u \, d\mathbf{X}_u$ takes values in $\mathbb{R}$. Let V be the path given by

$$V_t = \exp \left(\int_0^t K_u \, d\mathbf{X}_u - \frac{1}{2} \int_0^t (K_u \otimes K_u) \, d[\mathbf{X}]_u \right), \quad t \in [0, T].$$

Prove that V is the unique solution of the rough differential equation

$$V_t = 1 + \int_0^t V_u K_u \, d\mathbf{X}_u, \quad t \in [0, T].$$